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ORIGINAL ARTICLE

Type-I protein arginine methyltransferase inhibition primes anti-programmed cell death protein 1 immunotherapy in triple-negative breast cancer

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Abstract

Background: Immune-checkpoint blockade (ICB) therapy shows promise for treating aggressive triple-negative breast cancer (TNBC). However, only some patients benefit from ICB, revealing an urgent need for identifying novel strategies for sensitizing patients to ICB. Previously, the authors demonstrated that type-I protein arginine methyltransferases (PRMTs) regulated antiviral innate-immune responses in TNBC by altering RNA splicing. This study aimed to explore the effects of targeting type-I PRMTs on the tumor microenvironment (TME) and the efficacy of ICB therapy against TNBC.

Methods: Single-cell transcriptomic analysis was performed to investigate the effects of type-I PRMT inhibition on the TME, especially T-cell subsets. Single-cell T-cell receptor sequencing was performed to analyze the diversity and dynamics of the T-cell repertoire. A syngeneic murine model of TNBC was used to evaluate the therapeutic efficacy and immune memory effect of combining a type-I PRMT inhibitor (MS023) with an anti-programmed cell death protein 1 (PD-1) antibody.

Results: Type-I PRMT inhibition combined with anti–PD-1 therapy reduced tumor growth. Mechanistically, type-I PRMT inhibition reshaped the TME. Increased CD8 T-cell infiltration was verified using flow cytometry. Increased clonotypes and clonal diversity were also observed after MS023 treatment, which contributed to immune memory following combination treatment.

Conclusions: Targeting type-I PRMT can potentially improve immunotherapeutic efficacies in patients with TNBC. By enhancing the tumor immunogenicity and promoting a more favorable immune microenvironment, this combined approach may enable more patients with TNBC to benefit from immunotherapies.

KEYWORDS
immunotherapy, single-cell transcriptomic analysis, TNBC, type-I PRMT

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The first three authors contributed equally to this article.

INTRODUCTION

Triple-negative breast cancer (TNBC) is a highly malignant subtype of breast cancer that lacks estrogen receptor, progesterone receptor, and human epidermal growth factor receptor 2 (HER2) expression, which renders conventional therapies such as hormone therapy and HER2-targeted therapy ineffective. Therefore, an urgent need exists for new therapeutic strategies for TNBC.[1] Immunotherapy is a promising strategy for treating some solid tumors, including TNBC,[2] because TNBC exhibits higher tumor mutational burden (TMB) and greater immune infiltration than other breast cancer subtypes.[3–6] However, only a subset of TNBC patients derive significant clinical benefits from immunotherapy.[7] Certain chemotherapies (i.e., nab-paclitaxel) and targeted drugs (i.e., poly [ADP-ribose] polymerase inhibitors) can enhance the immunogenicity of tumor cells, thereby improving the efficacy of immunotherapies,[8–10] suggesting the potential for improving TNBC immunotherapies.

Aberrant alternative splicing is a hallmark of cancer.[11] Disrupting this process can activate viral-mimicry responses in cancer cells by promoting double-stranded RNA (dsRNA) accumulation,[12] which may augment immune infiltration within the tumor, enhancing the immune system's capacity to eliminate tumors. The pharmacological modulation of RNA splicing can foster the generation of immunogenic tumor peptides and amplify T-cell infiltration, eliciting an antitumor T-cell response.[13] These results highlight the potential of targeting aberrant alternative splicing for cancer treatment.

Type-I protein methyltransferases (PRMTs) catalyze the formation of asymmetric ω-N^G,N^G-dimethylarginine (ADMA) residues on target proteins and play a significant role in TNBC growth.[14] Asymmetric protein dimethylation affects various cellular processes including gene transcription, RNA splicing, DNA repair, and signal transduction.[15] Disrupting ADMA modification can elicit antitumor effects,[16] suggesting that targeting type-I PRMTs is promising for cancer therapy. Previously, we demonstrated that MS023 (a type-I PRMT inhibitor) disrupted RNA splicing and induced a viral-mimicry response by generating dsRNAs in cancer cells.[14] This study aimed to explore the effects of targeting type-I PRMTs on the tumor microenvironment (TME) and the efficacy of immune-checkpoint blockade (ICB) therapy against TNBC. Our findings may help establish novel strategies to benefit more patients with TNBC from immunotherapies.

MATERIALS AND METHODS

Animal experiments

Murine 4T1 TNBC cells were injected subcutaneously (2.0×10^5 cells) into 5- to 6-week-old female BALB/c mice. To assess the efficacy of an immune-checkpoint antibody, tumor-bearing BALB/c mice were randomly divided into four groups. *InVivoMAb*, an anti-mouse–PD-1 antibody (BioCell, West Lebanon, New Hampshire), was injected intraperitoneally every 3 days at a dose of 200 µg. MS023 (MedChemExpress, New Jersey) was injected intraperitoneally (60 mg/kg) every day. For tumor re-challenge experiments, murine 4T1 cells (3.5×10^5 cells) expressing luciferase were injected subcutaneously into two mice showing a complete response (CR) after 80 days and into three normal BALB/c mice. Imaging was performed to assess their tumor burdens 20 days after tumor implantation. Tumor volumes were calculated as follows: volume = length $\times$ width^2 $\times$ 0.5.

Single-cell RNA sequencing data analysis

After 2 weeks of treatment, the tumors of three mice per group were pooled together as one sample. The tumors were dissociated into single-cell suspensions. Approximately 8000 cells per group were used for single-cell RNA sequencing (scRNA-seq) library preparation using the Chromium Single-Cell 3′ Kit V3 (10X Genomics, California). The libraries were analyzed with the Illumina NovaSeq 6000 sequencing system (paired-end multiplexing, 150 base pairs). FASTQ files were converted to count matrices using CellRanger software (v7.0.0). The mm10 *Mus musculus* genome was used as a reference genome. The output was analyzed using Seurat software (v4.1.1) for dimensionality reduction and clustering.

Flow cytometry

At the end point of each animal experiment, tumors were collected, minced, and digested for 30 min at 37°C using Tumor Tissue Dissociation Solution (catalog no. SD004; Orgarid, Hunan, China) following the manufacturer's protocol to obtain single-cell suspensions. Then, each suspension was passed through a 40-µm cell strainer. The cells were incubated with the Live/Dead Cell Viability Reagent (catalog no. 565388; BD Pharmingen, New Jersey) for 30 min on ice and then washed with neutralizing phosphate-buffered saline containing 2% fetal bovine serum. For surface staining, cells were incubated with three different antibodies, including Ms CD45 (catalog no. 563891; BD Pharmingen), Ms CD3e (catalog no. 562600; BD Pharmingen), and Ms CD8a (catalog no. 553030; BD Pharmingen) at 4°C for 30 min. The samples were run on a CytoFLEX LX flow cytometer (Beckman Coulter, Brea, California), and compensation and gating were performed with FlowJo (v10.6.2).

Assessing T-cell receptor profiles

We performed T-cell receptor (TCR) sequencing and analysis using the Single Cell TCR V(D)J Enrichment Kit (10X Genomics) following the manufacturer's instructions. In addition, 5′-barcoded gene-expression libraries were sequenced using the Illumina HiSeq platform. The raw sequencing data were mapped to the mm10 *M. musculus* genome using CellRanger software (10X Genomics). The scRepertoire package (https://github.com/ncborcherding/scRepertoire; v1.7.0) of R software was used to analyze the T-cell repertoire diversity and dynamics.

Statistics analysis

All measurements were obtained from independent samples. *p* values were analyzed using an unpaired two-tailed *t*-test to study differences between two groups or one-way ANOVA among more than two groups. Statistical analyses were performed using GraphPad Prism (v8.0.2). Results were considered statistically significant at *p* value <.05.

RESULTS

MS023 synergizes with anti–PD-1 immunotherapy in suppressing tumor growth in vivo

Previously, we showed that type-I PRMT inhibition with MS023 altered mRNA splicing, leading to the expression of Alu sequences that can form cytosolic dsRNA.[14] This triggered an antiviral interferon (IFN)-mediated response in TNBC, which significantly enhanced tumor immunogenicity (Figure 1A). We hypothesized that targeting type-I PRMT could synergize with ICB. To evaluate the potential of targeting type-I PRMT for increasing the immunotherapeutic efficacy against TNBC, syngeneic BALB/c mice bearing murine TNBC cells (4T1) were treated with either MS023, anti–PD-1, or a combination of both (Figure 1B). Concurrent treatment with

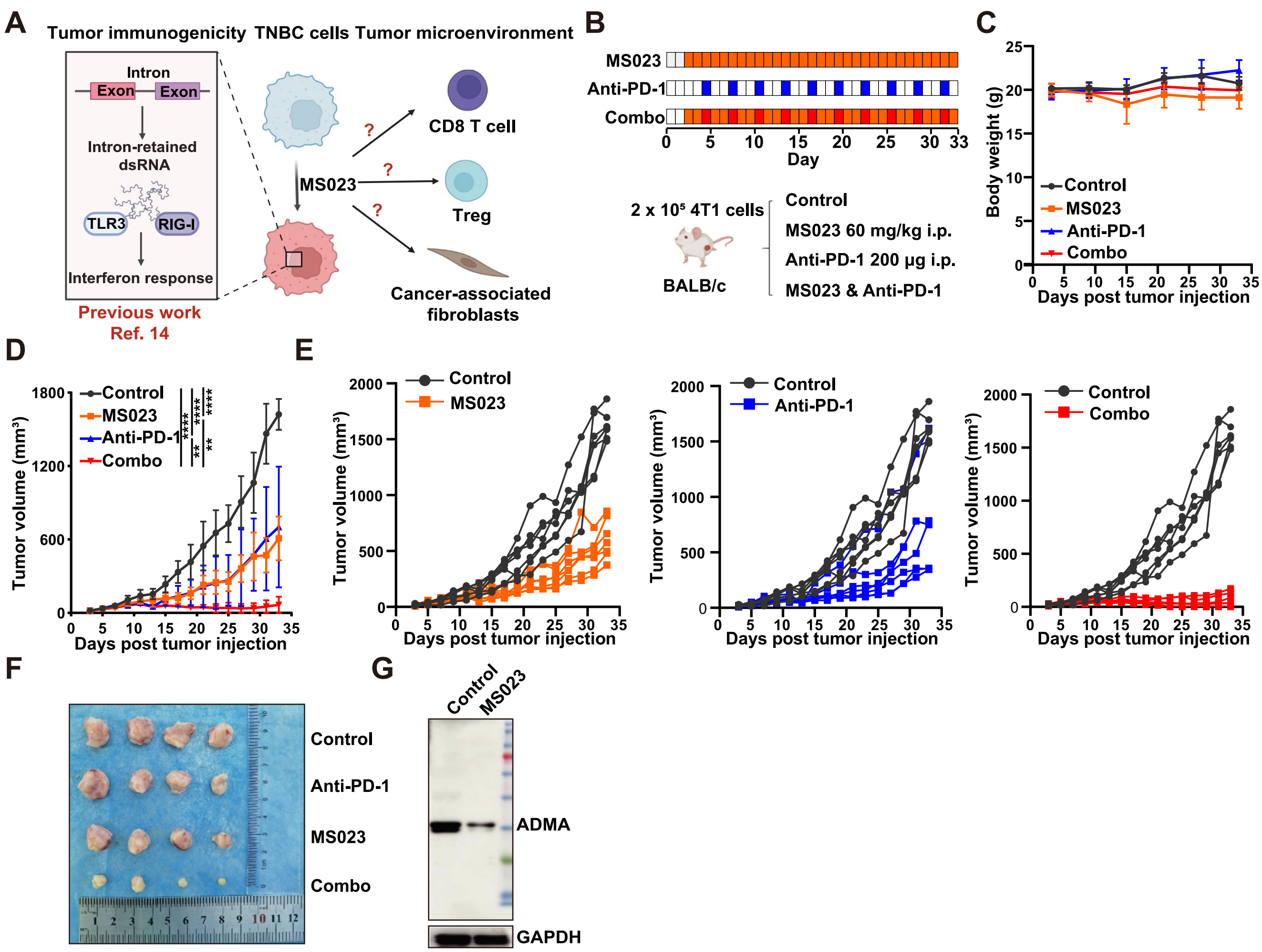

FIGURE 1 MS023 synergizes with anti–PD-1 immunotherapy in suppressing tumor growth in vivo. (A) Schematic diagrams illustrate the mechanisms through which MS023 regulates tumor cells (*left*, previous work) and the immune microenvironment (*right*, the focus of this study). (B) Treatment schema. (C) Body weights of BALB/c mice in the control, MS023, anti–PD-1, and combination-therapy groups ($n = 7$/group). The data is presented as the mean ± SD. (D) Xenograft tumor volumes in BALB/c mice administered 4T1 cells and the indicated treatments ($n = 7$/group). The data is presented as the mean ± SD. *****p* < .0001, ***p* < .01. (E) Growth curves of individual tumors from (D). (F) Representative ex vivo images of resected tumors from (D). (G) Immunoblot of tumors from mice treated with control or MS023 (at the experimental end point). ADMA indicates asymmetric dimethylarginine; anti–PD-1, anti-programmed cell death protein 1; SD, standard deviation.

MS023, anti–PD-1, or their combination demonstrated no toxic effects, as evidenced by body-weight measurements (Figure 1C). As expected, combining both regimens resulted in slower tumor progression than treatment with either regimen alone (Figure 1D,E). The tumor sizes in the combination-treatment group were substantially smaller than those in the other groups, as depicted in representative photographs (Figure 1F). Following MS023 treatment, the enzymes catalyzing asymmetric ADMA modifications of histone and nonhistone proteins (that serve as substrates for type-I PRMT) exhibited substantially lower expression levels (Figure 1G), validating the on-target effect of MS023 in vivo. Conclusively, combining the type-I PRMT inhibitor MS023 with anti–PD-1 immunotherapy synergistically enhanced the antitumor response in a mouse model of TNBC, suggesting it as a promising therapeutic approach for patients with TNBC.

MS023 reprograms the tumor immune microenvironment

To further investigate the mechanism underlying the sensitization to MS023-mediated immunotherapy, scRNA-seq was performed with tumor tissues from BALB/c mice exhibiting murine TNBC cells (4T1) treated with either normal saline or MS023. We identified six distinct cell populations, including cancer cells (*Krt10*, *Krt8*, and *Acta2*), T cells (*Cd3e*, *Cd3g*, and *Ncr1*), myeloid cells (*Csf1r*, *Cd83*, *Cd86*, *Itgam*, and *Itgax*), neutrophils (*Cd33* and *Csf3r*), cancer-associated fibroblasts (CAFs; *Col1a1*, *Col1a2*, *Col5a2*, *Pdpn*, and *Fgf7*), and endothelial cells (*Pecam1* and *Cd34*) (Figure 2A). Treatment with MS023 significantly increased the infiltration of immune cells (T cells, myeloid cells, and neutrophils) and decreased the proportion of cancer cells, CAFs, and endothelial cells, indicating that tumor immune microenvironment

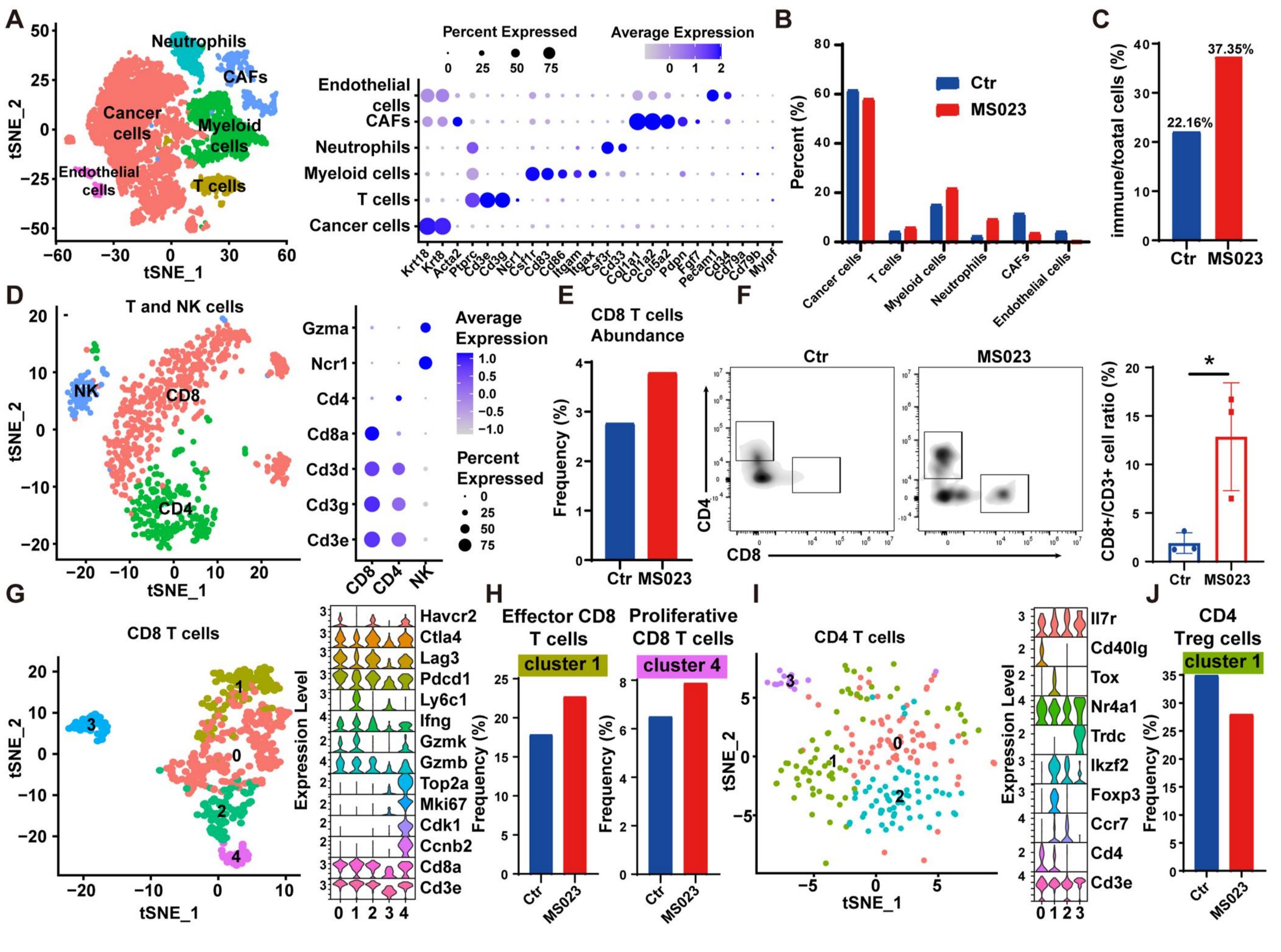

FIGURE 2 MS023 reprograms the tumor immune microenvironment. (A) tSNE plot showing cell populations identified within tumors from both the control and MS023 group (*left*). Markers used to distinguish the cell populations are shown in the dot plot (*right*). (B) Percentages of the different cell populations identified in (A). (C) Percentage of immune cells in each group. (D) tSNE plot of T-cell subclustering in 4T1 cell-derived tumors (*left*). The markers used are shown in the dot plot (*right*). (E) Proportions of CD8 T cells in each group. (F) Representative histograms showing flow cytometric analysis of the abundances of CD8 T cells in each group (*left*). Quantification of the frequencies of CD8 T cells in each group (*right*, $n = 3$/group). The data shown represent the mean ± SD. $^*p < .05$. (G) tSNE plot of CD8 T-cell subclusters present in 4T1 tumors (*left*). The markers used are shown in the violin plot (*right*). (H) Proportions of cells in CD8 T-cell subcluster 1 (*left*) and subcluster 4 (*right*) in each group. (I) tSNE of CD4 T-cell subclustering in 4T1 tumors (*left*). The markers used are shown in the violin plot (*right*). (J) Proportions of cells in CD4 T-cell subcluster 1 (Treg cells) in each group. tSNE indicates t-distributed stochastic neighbor embedding.

was reshaped (Figure 2B,C). Notably, enhancing immunogenicity can increase T-cell infiltration.[17] Furthermore, CAFs, the most abundant stromal cells in the tumor microenvironment (TME), are involved in the resistance to cancer therapeutics.[18]

To further investigate changes in T cells following MS023 treatment, we re-clustered the T-cell population and identified CD8, CD4, and natural killer (NK) cell subpopulations (Figure 2D). Flow cytometry analysis confirmed that the proportion of CD8 T cells was significantly higher in the MS023-treated group (Figure 2E,F). Considering that CD8 T cells are among the most potent effectors of anticancer immune responses, we subdivided them into five distinct subpopulations. (Figure 2G). Cluster 1 exhibited high expression levels of *Gzmb*, *Gzmk*, *Ifng*, and *Ly6c1*, which are characteristic markers of effector T cells. Cluster 4 represented highly proliferative T cells expressing *Ccnb2*, *Cdk1*, and *Mki67* (Figure 2G).[19] Following treatment with MS023, the proportions of cells in clusters 1 and 4 increased significantly (Figure 2H). We also investigated the role of CD4 T cells in the antitumor immune response and re-clustered the CD4 T-cell population into four subpopulations (Figure 2I). Cluster 1, characterized by high *Foxp3* expression, was identified as regulatory T (Treg) cells, which suppress antitumor immune responses through various mechanisms.[20] The proportion of Treg cells was significantly lower in the MS023-treated group. These results suggest that MS023 treatment reshaped the tumor immune microenvironment and promoted the infiltration of cytotoxic immune cells in our murine model of TNBC.

T-cell immune repertoire changes following MS023 treatment

Through scRNA-seq, we demonstrated that type-I PRMT inhibition significantly enhanced the infiltration of cytotoxic T cells in tumors. However, the alterations in the dynamics and diversity of the T-cell immune repertoire following MS023 treatment remained unclear. Therefore, we performed single-cell T-cell receptor (scTCR) sequencing. The T-cell diversity, estimated based on complementarity-determining region 3 (CDR3) and gene sequences, revealed significantly more unique clonotypes in the MS023-treated group than in the control group (Figure 3A). This finding suggests that MS023 treatment promoted the generation of activated T cells that could recognize new tumor-immunogenic antigens. The length of CDR3, another crucial determinant of the T-cell repertoire diversity, increased after MS023 treatment (Figure 3B). Longer CDR3 sequences (25–30 residues long) were previously observed to be better suited for recognizing certain antigenic epitopes compared to the more common shorter CDR sequences.[21] Moreover, the clonal diversity increased significantly following MS023 treatment (Figure 3C). To gain a deeper understanding of the TCR repertoire changes that occurred in the T-cell subpopulations, we performed an integrated scTCR and scRNA sequencing analysis. Interestingly, the clonotype frequencies were higher in CD8 T cells than in CD4 T cells, and no TCRs were detected in NK cells, as expected (Figure 3D). We further analyzed the unique clonotypes in different CD8 subpopulations and found that they were most prevalent in cluster 4 (Figure 3E). Moreover, the clonal diversity of cluster 4 was highest among all the subpopulations (Figure 3F). Cluster 4 was identified as a proliferating cell cluster with the highest metrics in terms of the clone repertoire. Consequently, we hypothesized the existence of differentiation pathways between these cell clusters and conducted a pseudotime analysis to test this hypothesis. As anticipated, cluster 4 was positioned at the starting point of the pseudotime.

MS023 enhances immune memory generation when administered in combination with anti–PD-1 therapy

MS023 can enhance T-cell infiltration and TCR expansion. Therefore, we hypothesized that MS023 may contribute to immune memory generation. To test this hypothesis, we selected mice that exhibited complete tumor regression after receiving combination therapy and being inoculated with 4T1 tumor cells, as well as three control mice that were never inoculated with tumor cells. We then re-inoculated these mice with tumor cells (Figure 4A). Tumor cell proliferation was significantly lower in the CR group than in the naive group (Figure 4B). In vivo bioluminescence analysis further confirmed that the bioluminescence intensity of the primary tumors was significantly lower, along with the occurrence of lung metastases in the CR group (Figure 4C). These results suggest that combination therapy with MS023 and anti–PD-1 can promote the production of immune memory cells capable of maintaining an immune response against tumors.

DISCUSSION

We demonstrated that MS023, a type-I PRMT inhibitor, synergized with anti–PD-1 immunotherapy to enhance antitumor responses in a mouse model of TNBC. Furthermore, MS023 treatment reshaped the tumor immune microenvironment and promoted the infiltration of cytotoxic immune cells while decreasing the proportion of regulatory T cells. MS023 treatment also promoted the generation of activated T cells that can recognize new tumor-immunogenic antigens.

PRMTs are crucial epigenetic modifiers that catalyze the methylation of arginine residues in proteins.[22] Mounting evidence suggests that genetically or pharmacologically targeting PRMTs, such as the type-I PRMT, (CARM1),[23] type II PRMT (PRMT5),[24] and type III PRMT (PRMT7),[25] can potentially enhance tumor immunogenicity, ultimately improving the efficacy of ICB in melanoma or colorectal cancer. However, the precise role of PRMTs in regulating antitumor immunity in TNBC remains elusive. Previously, we showed that a type-I PRMT inhibitor stimulated intron-retained dsRNA formation and interferon-response signals,[14] which prompted us to investigate the efficacy of combining a type-I PRMT inhibitor with anti–PD-1 therapy in this study. MS023 treatment effectively reshaped the immune microenvironment, resulting in enhanced antitumor immune

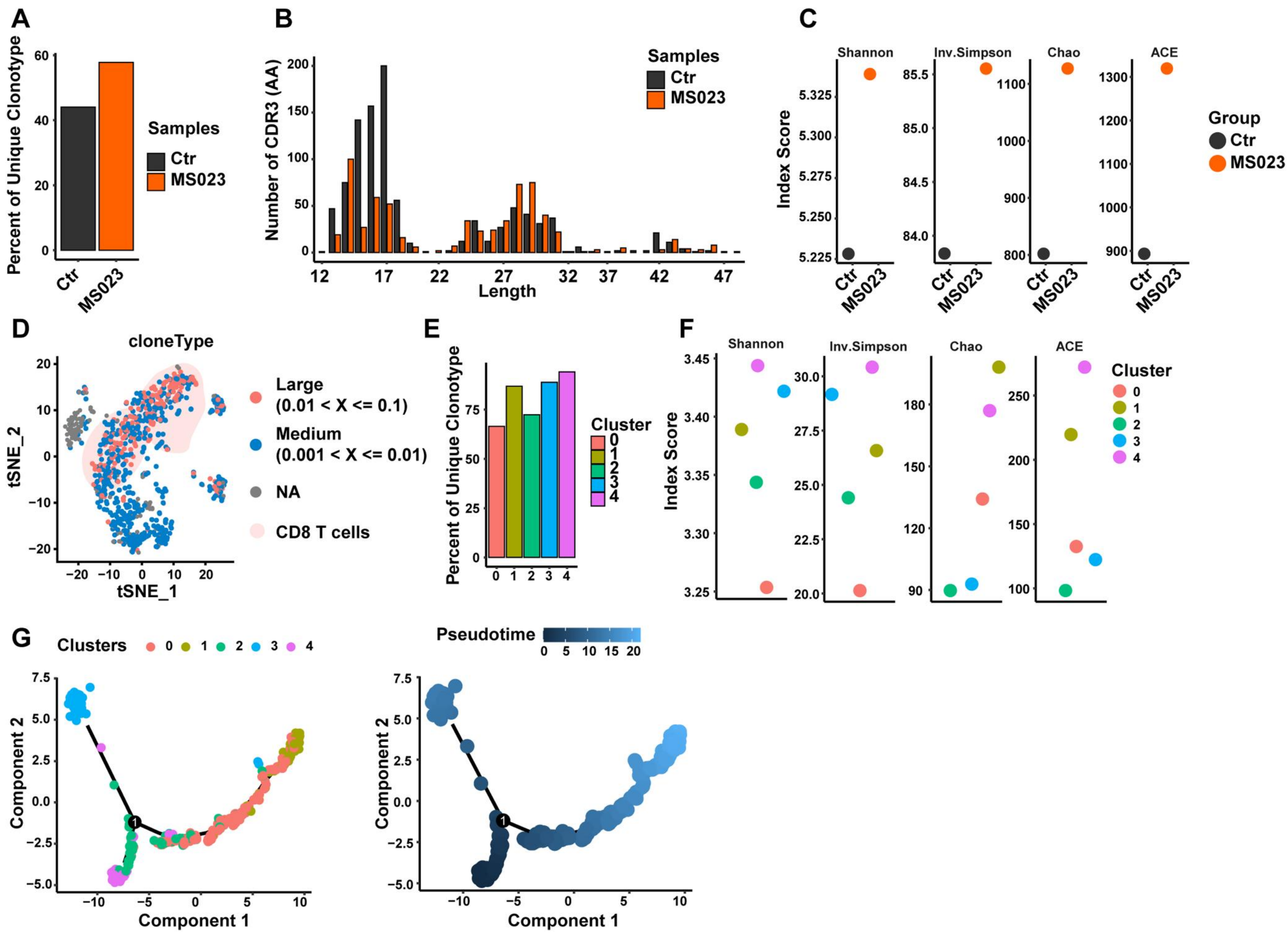

FIGURE 3 T-cell immune repertoire changes following MS023 treatment. (A) Percentages of unique clonotypes among all clonotypes in each group. (B) Distribution of the CDR3 nucleotide length in each group. (C) Diversity analysis using the Shannon, inverse Simpson, Chao, and ACE indices for each group. (D) Individual cells are grouped according to the number of clonotypes and are overlaid with the tSNE projection in Figure 2D. (E) Percentages of unique clonotypes among all clonotypes in each CD8 T-cell subcluster in each group. (F) Diversity analysis in each CD8 T-cell subcluster in each group. (G) Pseudotime trajectory of the CD8 T-cell population. ACE indicates abundance-based coverage estimator.

responses and increased sensitivity to anti–PD-1 therapy. However, the specific contributions of MS023-induced activation of the interferon-signaling pathway in tumor cells to immune microenvironment remodeling and anti–PD-1 therapy sensitization should be explored further.

Compared with many other PRMT-targeting therapies that primarily enhance CD8 T-cell infiltration, MS023 treatment induced notable alterations in the dynamics and diversity of the T-cell immune repertoire. Furthermore, MS023 treatment substantially reduced the abundance of CAF cells, which potentially contributed to the increased sensitivity of PD-1 blockade when combined with MS023. However, the related role and impact of CAFs should be further investigated.

In addition to regulating tumor immunogenicity, PRMTs also play pivotal roles in immune cell development and maintenance. For instance, PRMT1 affected the functions of Treg cells through asymmetric FOXP3 methylation at R48 and R51.[26] Our findings revealed that MS023 treatment increased the expression of functional markers in CD8 T cells, which loosely reflected their enhanced activity. However, the molecular mechanisms by which type-I PRMTs regulate the function of CD8 T cells warrant further investigation.

Although type-I PRMTs are promising cancer-therapy targets, clinical data obtained by targeting type-I PRMTs with GSK3368715 for tumor treatment revealed a higher incidence of thromboembolic events during the dose-escalation phase.[27] Treatment with MS023 demonstrated no significant side effects, as evidenced by body-weight measurements and results from hematoxylin–eosin staining (Figure S1). Furthermore, some new small-molecule inhibitors targeting PRMTs with potent antitumor activity and minimal side effects are being continuously developed currently.[28] The use of a combination therapy strategy also has the potential to reduce dosages. Therefore, our findings offer the prospect of minimizing side effects through the combination of immunotherapy.

MS023 treatment significantly increased the number of unique clonotypes and the length of CDR3, suggesting that MS023 treatment might help remodel the TCR repertoire. However, the targets of

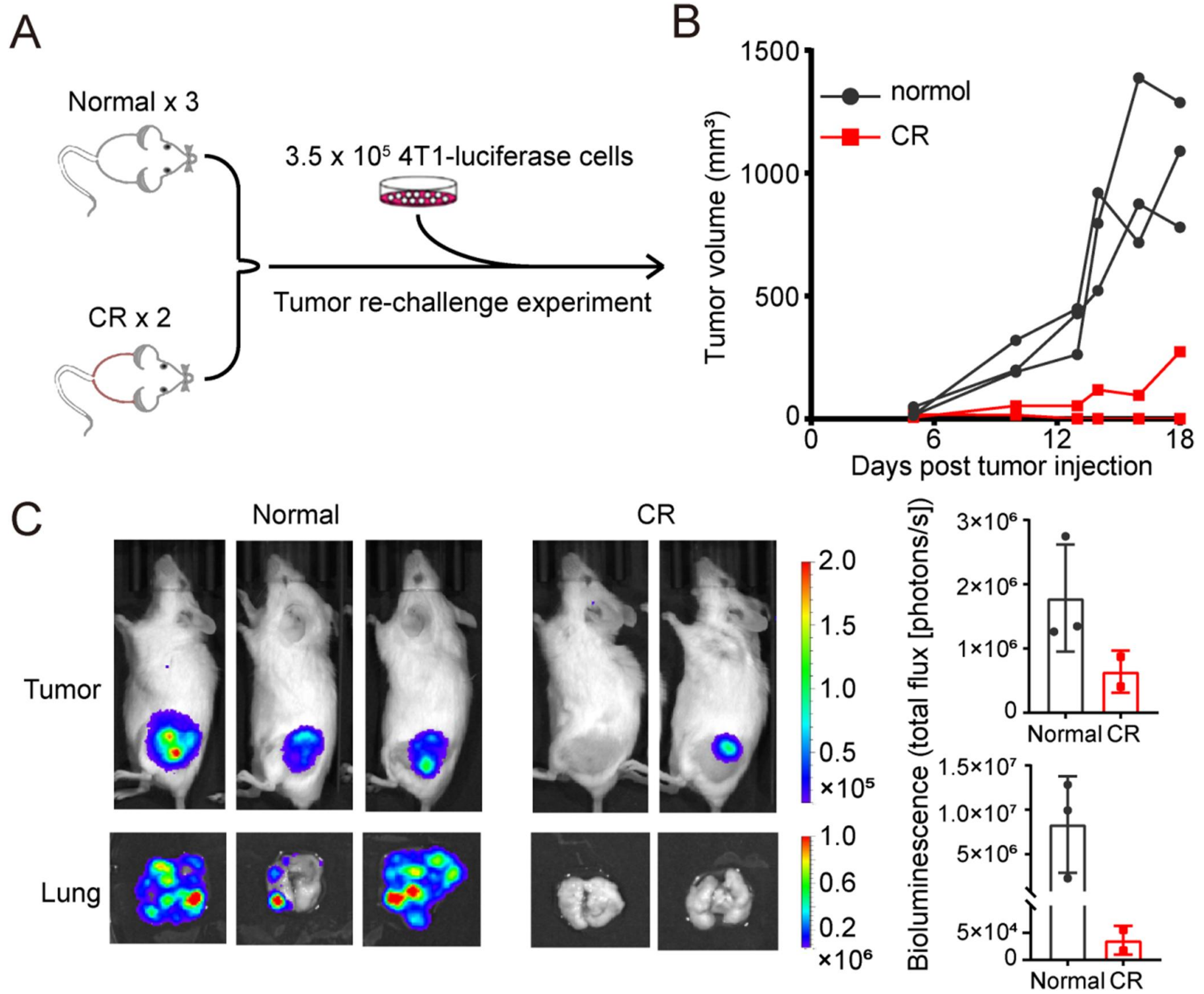

FIGURE 4 MS023 enhances immune memory generation in combination with anti–PD-1 therapy. (A) Tumor re-challenge schema. (B) Growth curves of individual tumors in the normal ($n = 3$) and CR ($n = 2$) groups. (C) Bioluminescence images of mice and ex vivo bioluminescence images of lung metastases in each group (*left*). Bioluminescence signal for tumors and lung metastases in each group (*right*). anti–PD-1 indicates anti-programmed cell death protein 1; CR, complete response.

MS023 (type-I PRMTs) participate in both arginine methylation and RNA splicing.[29] Therefore, the increased length of CDR3 sequences might be attributable to alterations in alternative splicing. The precise mechanisms underlying these changes and their implications for TCR repertoire remodeling require further investigation.

We made several promising observations in this study. MS023 treatment enhanced T-cell infiltration and TCR expansion, suggesting that MS023 potentially contributed to the generation of activated T cells that can recognize new tumor-immunogenic antigens. The increase in the number of unique clonotypes and the length of CDR3 was revealed by our in-depth analysis of the T-cell immune repertoire using scTCR sequencing.

In our discussion of the potential role of MS023 in promoting the generation of T cells capable of recognizing new antigens, we have considered the interplay between MS023, type-I PRMTs, and alternative-splicing events. Type-I PRMTs can participate in alternative RNA splicing, and aberrant alternative splicing can potentially promote the generation of tumor neoantigens.[30] Inhibiting type-I PRMTs with MS023 caused changes in alternative splicing in tumor cells, implying that MS023 might facilitate the production of neoantigens. Our findings indicate that MS023 treatment led to TCR repertoire remodeling, suggesting that MS023 might promote the generation of T cells that recognize new tumor antigens.

Considering the crucial role of type-I PRMTs in reshaping the TME and modulating immune responses, future research should elucidate the impact of inhibiting type-I PRMTs on immune activation in various cancers. Such a broader investigation would support more generalized conclusions regarding the enhancement of ICB efficacy through type-I PRMT inhibition. Further study is needed to assess whether MS023 can demonstrate varying sensitization effects on immune-checkpoint inhibitors, particularly those with distinct molecular mechanisms involved in T-cell activation.

In conclusion, our findings have demonstrated the synergistic effect of type I PRMT inhibition in conjunction with anti–PD-1 immunotherapy, resulting in the robust suppression of tumor growth in vivo. Furthermore, a comprehensive single-cell analysis has shed light on the profound impact of MS023 on T-cell infiltration, the dynamics of immune repertoire, and the generation of immune memory. This comprehensive understanding of the intricate interplay between different T-cell subpopulations and their roles in tumor eradication holds the potential to pave the way for the development of innovative immunotherapeutic strategies for improving cancer-treatment outcomes.

AUTHOR CONTRIBUTIONS

Qin Wu: Conceptualization, methodology, and writing–original draft. **Sheyu Zhang**: Methodology, experimentation, and writing–original draft. **Lu Guo**: Data analysis, interpretation, and writing–original draft. **Ziwen Zhang**: Methodology and experimentation. **Xueying Liu**: Experimentation. **Wenjun Chen**: Writing–reviewing and editing. **Yong Wei**: Supervision and writing–reviewing and editing. **Xiaojia Wang**: Supervision and project administration. All authors approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

All authors report no conflicts of interest.

DATA AVAILABILITY STATEMENT

Raw sequencing reads of all single-cell experiments (scRNA-seq and scTCR-seq) are available to researchers on request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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